

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

I Semester of B. Sc. Backlog Examination , March, 2018

Course code & Name BS 108 & Biochemistry

Date: 26/03/2018 Day: Monday Time: 10.00 AM To 10.30 AM Maximum Marks: 20

MCQ

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Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I	Choose the correct answer for the following questions.	20
1.	The general formula of carbohydrate is	01
	(i) $(\text{CH}_2\text{O})_n$	(ii) $(\text{CHO})_n$
	(iii) $(\text{C}_4\text{H}_2\text{O})_n$	(iv) $(\text{C}_2\text{H}_2\text{O})_n \text{COOH}$
2.	DNA consists of two antiparallel strands of nucleotide chains held together by	01
	(i) peptide bonds.	(ii) polar covalent bonds.
	(iii) hydrogen bonds.	(iv) complementary base pairing bonds.
3.	In polysaccharides, monosaccharides are joined by	01
	(i) Peptide bond	(ii) covalent bond
	(iii) glucose bond	(iv) Glycosidic bond
4.	The naturally occurring form of amino acid in proteins	01
	(i) L-amino acids only	(ii) D-amino acids only
	(iii) Both L & D amino acids	(iv) None of these
5.	Nucleotides have phosphates attached at the ____ carbon atom of the sugar.	01
	(i) 8'	(ii) 5'
	(iii) 3'	(iv) 1'

6.	Amino acids are transported to the ribosome for use in building the polypeptide by		01
	(i) mRNA molecules.	(ii) tRNA molecules.	
	(iii) rRNA molecules.	(iv) DNA polymerase molecules.	
7.	Enantiomers are		01
	(i) a pair of compounds that are non superimposable mirror-images of one	(ii) molecules with both hydrophobic and hydrophilic characteristics	
	(iii) not chiral	(iv) compounds with very different chemical composition	
8.	Raffinose is a		01
	(i) Monosaccharide	(ii) Oligosaccharide	
	(iii) Trisaccharide	(iv) Polysaccharide	
9.	Glycogen in animals are stored in		01
	(i) liver and stomach	(ii) liver and adipose tissue	
	(iii) liver and bile	(iv) liver and muscles	
10.	To possess optical activity, a compound must be		01
	(i) a carbohydrate	(ii) a hexose	
	(iii) D-glucose	(iv) asymmetric	
11.	A common function of Thiamin, Riboflavin and Niacin is that:		01
	(i) They all are used in synthesis of blood clotting proteins	(ii) They all work as a part of a coenzyme used in energy metabolism	
	(iii) They all help to strengthen blood vessel walls	(iv) They are used to stabilize cell membranes	
12.	Which one of these vitamins is involved in blood clotting?		01
	(i) Vitamin B ₆	(ii) Vitamin B ₁₂	
	(iii) Vitamin E	(iv) Vitamin K	
13.	Which one of these vitamins is involved in calcium homeostasis?		01
	(i) Vitamin A	(ii) Vitamin D	
	(iii) Vitamin E	(iv) Vitamin K	
14.	Which of the following groups of elements represent the most common elements in biological systems?		01
	(i) Carbon, nitrogen, sodium, and hydrogen.	(ii) Carbon, nitrogen, oxygen, and hydrogen.	
	(iii) Carbon, sodium, potassium, and nitrogen.	(iv) Carbon, sodium, hydrogen, and oxygen.	
15.	Which of the following is not a macromolecule?		01
	(i) Water	(ii) Proteins	
	(iii) Carbohydrates	(iv) Lipids	
16.	A common lipid for energy storage is		01
	(i) phospholipid.	(ii) steroid.	
	(iii) triglycerides.	(iv) cholesterol.	
17.	The nitrogen base not found in DNA is called		01
	(i) uracil.	(ii) adenine.	
	(iii) thymine.	(iv) cytosine.	

18.	Absorbance at 280 nm exhibited by protein is due to		01
	(i) aromatic amino acids	(ii) aliphatic amino acids	
	(iii) all amino acids	(iv) Non-polar amino acids	
19.	Humans are unable to digest		01
	(i) starch	(ii) complex carbohydrates	
	(iii) denatured proteins	(iv) cellulose	
20.	The simplest amino acid is		01
	(i) glycine	(ii) proline	
	(iii) tyrosine	(iv) lysine	

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

I Semester of B. Sc. Backlog Examination, March, 2018

Course code & Name BS 108 & Biochemistry

Date: 26/03/2018 Day: Monday Time: 10.31 AM To 1.00 PM Maximum Marks: 50

Instructions:

1. Section I and II must be attempted in TWO ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION -I		
Q - II Answer the following questions as directed (Any 5)		20
1.	What is mutarotation? Give example.	04
2.	Methanol is highly toxic because it is converted to formaldehyde in a reaction catalyzed by the enzyme alcohol dehydrogenase ($\text{NAD}^+ + \text{methanol} \rightarrow \text{NADH} + \text{H}^+ + \text{formaldehyde}$). Large amount of ethanol enough to cause intoxication under normal circumstance is administered as a part of the medical treatment for methanol poisoning. Explain this in terms of what you know about this enzymatic reaction.	04
3.	a) What are hexoses? What are some examples of hexoses with important biological functions? b) What are pentoses? What are the roles of pentoses in DNA and RNA molecules?	2+2
4.	What is the importance of the primary structure of a protein? What is the difference between alpha-helix and beta-sheet protein conformations?	2+2
5.	What are antioxidants? Which vitamins can act as antioxidants? What are the signs and symptoms of vitamin E deficiency?	2+2
6.	What are the functions of carbohydrate?	04
7.	Write short notes on Hydrogen bond directionality and strength.	04

Q-III	Answer the following questions as directed (Any 10)	30
1.	What is the relationship between vitamin D and sunlight? What disease is caused by vitamin D deficiency? Which tissue does it affect?	1x3
2.	What vitamins make up the B complex? What problems are caused by a lack of these vitamins?	1+2
3.	What are the three main types of RNA and what are their functions?	03
4.	What is the complementary sequence of nitrogenous bases for an AGCCGTTAAC fragment of a DNA chain? Is there any situation in which DNA is made based on an RNA template? Which is the enzyme involved?	1x3
5.	Biochemical tests for qualitative detection of carbohydrates.	03
6.	What are the medical and biochemical importance of water?	03
7.	Write short notes on Artificial Sweeteners	03
8.	How peptide bond is formed?	03
9.	Write a short note on Sugar alcohol.	03
10.	What are essential fatty acids? Give names of 3 essential fatty acids? What are the characteristics of sterol?	1x3
11.	What is the rule for the pairing of nitrogenous bases within the DNA molecule? What about in RNA molecules? Is this last question relevant?	1x3
12.	What is the role of vitamin C in the body? What damage is caused by a lack of vitamin C? Why was this deficiency also known as “sailors' disease”?	1x3
13.	How the Body regulates Blood Sugar level?	03